

Shoujing, Guo (1231–1316) Chinese engineer astronomer, and mathematician, known for his Shoushi Calendar, which accurately presented 365 days in a year. Shoujing also invented an astrological compass, built hydraulic clocks, engineered the Kunming Lake reservoir in Beijing and developed spherical trigonometry.

Siemens, Werner von (1816–92) German electrical engineer remembered for his role in developing the telegraph industry. Inventor of the electric dynamo and an electroplating process, Siemens laid the first telegraph line in Germany and co-founded the telegraph company that is now called Siemens AG. The unit of electrical conductance bears his name.

Smith, William (1769–1839) English geologist and engineer who founded the science of stratigraphy. While working as a canal-site surveyor across Britain, Smith studied regional rock strata and the fossils within each layer, which enabled him to establish geological time periods. Smith produced the first geological map of England and Wales, in 1815.

Snell, Willebrord (1580–1626) Dutch physicist and mathematician credited with discovering the law of refraction. In 1617, he presented a method of measuring Earth by triangulation, and in 1621, developed his law of refraction.

Snow, John (1813–58) English physician and pioneer of modern epidemiology. Snow is best known for showing cholera to be a waterborne disease—a theory he published in 1839 and confirmed in 1854 through his investigation of London's Broad Street pump outbreak. He also promoted gaseous anesthesia after administering chloroform to Queen Victoria.

Somerville, Mary (1780–1872) Scottish astronomer, geographer, and popularizer of science. With little formal education, Somerville won acclaim for her 1831 translation of Pierre-Simon Laplace's *The Mechanism of the Heavens*. Celebrated as the "Queen of the Sciences" for her numerous and wide-ranging books, in 1835, she and Caroline Herschel became the first women members of the Royal Astronomical Society.

Sørensen, Søren Peder Lauritz (1868–1939) Danish biochemist famed for introducing the pH scale for expressing hydrogen ion concentration as a measure of acidity. Sørensen also contributed to the chemical technology of Denmark's spirits and explosives industries.

Spallanzani, Lazzaro (1729–99) Italian biologist and physiologist noted for his experimental research on animal reproduction and bodily functions. Spallanzani discredited the theory of spontaneous generation and showed that living cells use oxygen and give off carbon dioxide. He also proved that mammal reproduction requires semen and an ovum, and was the first to artificially inseminate a dog.

Spitzer, Lyman (1914–97) American theoretical physicist and astronomer who made significant contributions to the study of interstellar matter, plasma physics, and the dynamics of star clusters. Spitzer's 1946 proposal of a space telescope led to the development of the Hubble telescope, and he helped design the ultraviolet astronomy satellite, Copernicus.

Stahl, Georg (1660–1734) German physician and chemist who founded the phlogiston theory of combustion, which states that all substances that burn contain a substance called phlogiston.

Although later discarded, this theory was accepted for decades because it was very useful, especially in the mining industry.

Swammerdam, Jan (1637–80) Dutch microscopist who helped found the fields of comparative anatomy and entomology. After designing a dissecting microscope, he recorded his observations of the structure, classification of insects, and metamorphosis. The first to describe red blood cells, he also discovered Swammerdam valves in the lymphatic vessels.

Swan, Joseph (1828–1914) English physicist and chemist whose early incandescent light bulb predated that of Thomas Edison. While an assistant at a chemical manufacturing firm, Swan made important contributions to photography, and later legal disputes between Swan and Edison over the light bulb led to the Edison & Swan United Electric Light Company partnership.

Talbot, William Henry Fox (1800–77) English chemist and photography pioneer. Talbot is best known for his calotype photographic process that produced negatives from which prints could be taken. Talbot had 12 patents to his name and published over 50 articles on mathematics, astronomy, and physics. His book, *The Pencil of Nature*, was the first to feature photographic illustrations.

Tansley, Arthur (1871–1955) English ecologist and conservationist who coined the term ecosystem. Tansley advocated the study of plants within their natural communities—an approach central to modern ecology. Tansley's best-known book was the 1939 *The British Islands and their Vegetation*.

Tesla, Nikola (1856–1943) Serbian engineer and pioneering inventor in the fields of electricity and radio transmission. Tesla emigrated to America in 1884 where he worked with Thomas Edison and sold patents to George Westinghouse. He invented the Tesla coil transformer, the induction motor, and discovered the rotating magnetic field. The unit of magnetic induction bears his name.

Thomson, Joseph John (1856–1940) English physicist who discovered the electron. A professor of experimental physics at Cambridge University, Thomson also developed the mathematical theory of electricity and magnetism, discovered the natural radioactivity of potassium, and invented the mass spectrometer. He received the 1906 Nobel Prize in Physics for his study of conduction of electricity through gases.

Thomson, William (Lord Kelvin) (1824–1907) Scottish physicist and thermodynamic pioneer known for determining the correct value of absolute zero. Appointed a professor at Glasgow University at 22, Thomson developed the second law of thermodynamics and the electromagnetic theory of light, determined the value of absolute zero, and helped lay the first transatlantic telegraph cables. Deeply religious, he used his estimate of Earth's age to argue against evolution by natural selection.

Trevithick, Richard (1771–1833) English engineer and inventor of the first successful steam-powered railroad locomotive. Trevithick's engines were first used to power stationary mill and mine machinery, but he invented the first self-propelled road vehicle in 1801, and the railroad locomotive in 1803.

Tull, Jethro, English agronomist and inventor (1674–1741) See p.126

Turing, Alan (1912–54) English mathematician widely regarded as the father of computer science. During World War II, Turing developed the "Bombe," a prototype for electronic computers, which helped to crack the German enigma code. His development of the theoretical Turing machine, the Automatic Computing Engine, and the Ferranti Mark I paved the way for modern computing.

Venter, Craig, American biologist (1946–) See p.347

Vesalius, Andreas (1514–64) Flemish physician and founder of modern anatomy at Padua University. Vesalius's dissections of human cadavers informed his *The Seven Books on the Structure of the Human Body*, which included many detailed illustrations of internal human anatomy. He revolutionized anatomical teaching by insisting on the importance of close observation and the use of human cadavers.

Villasante, Manuel Losada (1929–) Spanish biologist and biochemist noted for his research on the photosynthetic assimilation of nitrogen. Losada's work focuses on biochemical and biological systems that can transform solar energy into chemical energy.

Virchow, Rudolf Carl (1821–1902) German physician and founding figure of modern pathology. Virchow popularized the expression "every cell is derived from a cell" and showed that disease occurs as a result of changes in normal cells. A pioneer of social medicine, he advocated the advancement of public health.

Volta, Alessandro (1745–1827) Italian physicist and inventor of the voltaic pile—the first electric battery to produce an electric current. A professor of physics at Pavia University, Italy, Volta also invented the electrophorus—a device that produced a static electric charge, and was the first to isolate methane gas. The unit of electrical potential—the volt—bears his name.

Vries, Hugo de (1848–1935) Dutch botanist known for his research into the nature of mutations in plant breeding. While a professor at Amsterdam University, Vries coined the terms "mutation," "isotonic," and "pangene" (later shortened to gene), and rediscovered Gregor Mendel's laws of heredity, while being unaware of Mendel's work at the time.

Waals, Johannes Diderik van der (1837–1923) Dutch physicist awarded the 1910 Nobel Prize in Physics for his equation of state for gases and liquids. The equation explained why real fluids do not obey the ideal gas laws at high pressures. His work led to the liquefaction of hydrogen and helium, and the study of temperatures near absolute zero.

Wallace, Alfred Russel, British naturalist (1823–1913) See p.203

Warburg, Otto Heinrich (1883–1970) German biochemist and physician awarded the 1931 Nobel Prize in Physiology or Medicine for his research on cancerous tumors and the respiration of cells. Director of the Kaiser Wilhelm Institute, Warburg discovered the nature and mode of action of the respiratory yellow enzyme, outlining his research in *The Metabolism of Tumors* (1931).

Watson, James Dewey (1928–) American geneticist and co-discoverer of the double helix structure of deoxyribonucleic acid (DNA). Watson shared the 1962 Nobel Prize in Physiology or Medicine with Francis Crick and Maurice Wilkins for the discovery. Watson became Director of the Cold Spring Harbor Laboratory, and took a leading role in the Human Genome Project.

Watt, James, British engineer and inventor (1736–1819) See p.150

Wegener, Alfred, German geophysicist and meteorologist (1880–1930) See p.252

Weinberg, Steven (1933–) American physicist and co-recipient of the 1979 Nobel Prize in Physics for his work in formulating the electroweak theory. Weinberg's theory, published in his 1967 article *A Model of Leptons* explains that electromagnetic and weak forces are indistinguishable at extremely high temperatures, such as those occurring during the Big Bang.

Weismann, August (1834–1914) German biologist and founder of the modern science of genetics. Weismann is known for his germ plasm theory, which states that all living things are born with a special and stable hereditary substance. A supporter of Darwin, Weismann opposed the idea of the inheritance of acquired characters.

White, Gilbert (1720–93) English naturalist, clergyman and author of *The Natural History and Antiquities of Selborne*. White's observation journals on his garden have attained English classic status.

Wöhler, Friedrich (1800–82) German chemist who became the first to synthesize an organic compound, urea, from an inorganic substance, ammonium cyanate. Professor of chemistry at Göttingen University, Germany, Wöhler also discovered calcium carbide, isolated the elements silicon and beryllium, and developed a way of preparing metallic aluminum.

Yalow, Rosalyn Sussman (1921–2011) American medical physicist and co-recipient of the 1977 Nobel Prize in Physiology or Medicine for developing radioimmunoassay. This is a method of measuring minute substances in the blood, such as hormones, enzymes, and vitamins. Director of the Berson Research Laboratory, Yalow was the second woman to win a Nobel Prize in this field.

Yukawa, Hideki (1907–81) Japanese physicist awarded the 1949 Nobel Prize in Physics for his theory of elementary particles. Yukawa's prediction of the existence of the meson, a subatomic particle hundred times heavier than the electron, would inform later research on nuclear and high-energy physics. He joined other scientists in signing the Russell–Einstein Manifesto for nuclear disarmament in 1955.

Zhang Heng (78–139) Chinese geographer, mathematician, and astronomer who invented a device to record any earthquake within 310 miles (500 km) by causing a ball to drop from a model dragon into a frog's mouth below, thus making a sound. Zhang also calculated the value of pi, created a comprehensive star map, and explained solar and lunar eclipses.